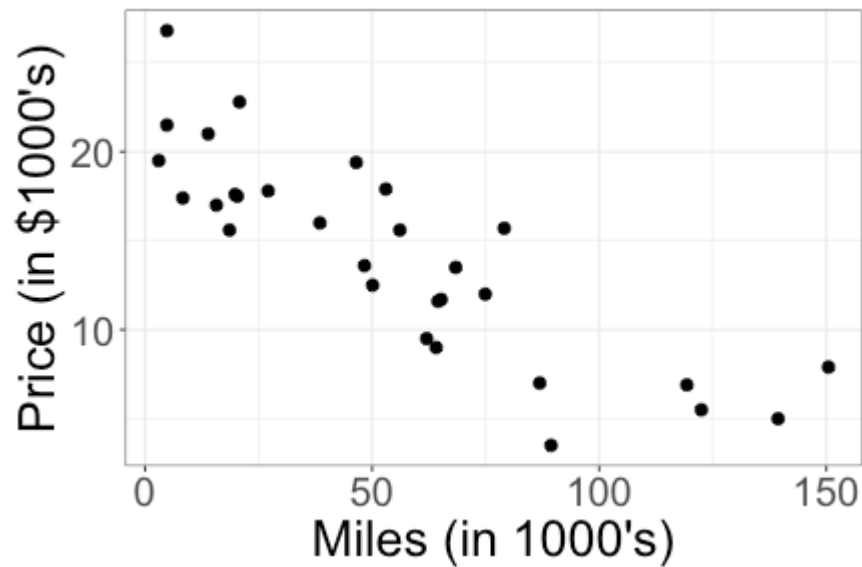


Inference for linear regression slope

Inference

Is there a relationship between mileage and price for used Honda Accords?

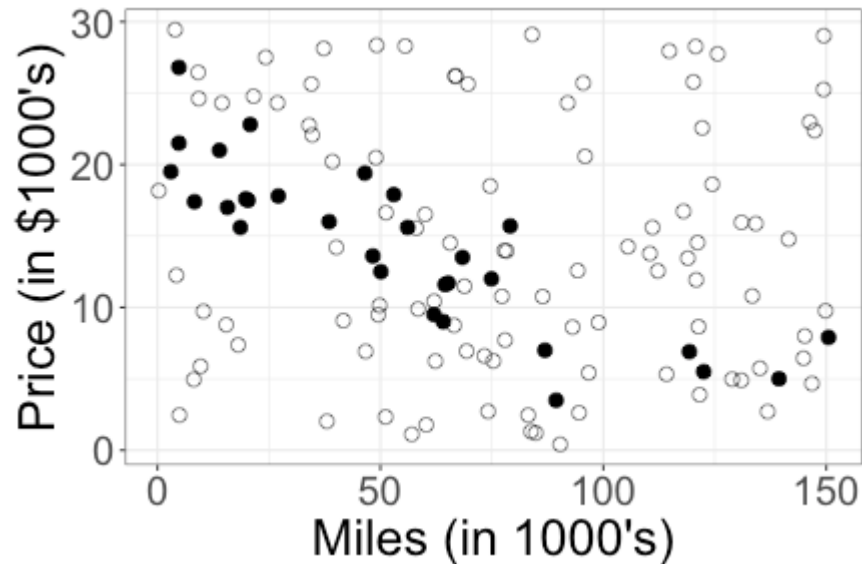


$$\hat{\beta}_1 = -0.12$$

It looks like it, but is that just a fluke?

Is the observed relationship a fluke?

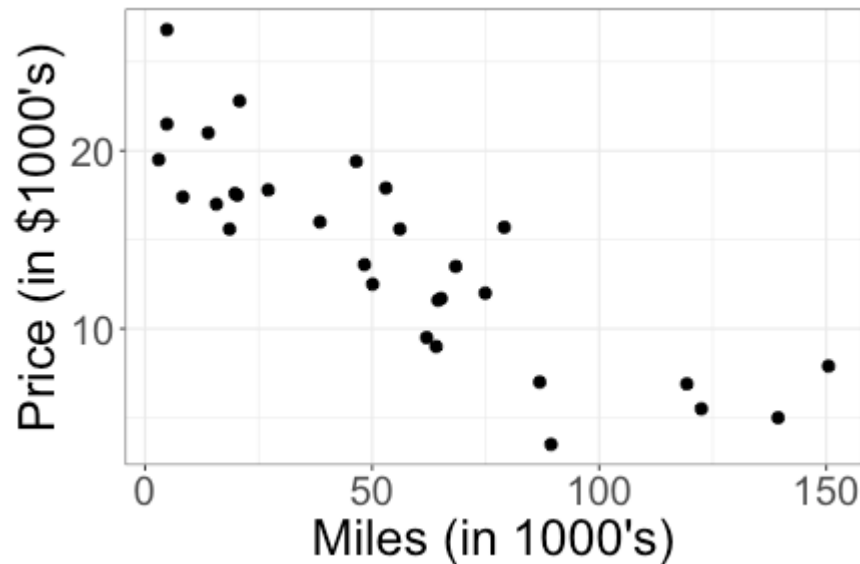
What if the population scatterplot looks like this?



Translating the question

if there is no
relationship,
then $\beta_1 = 0$

Is there a relationship between mileage and price?

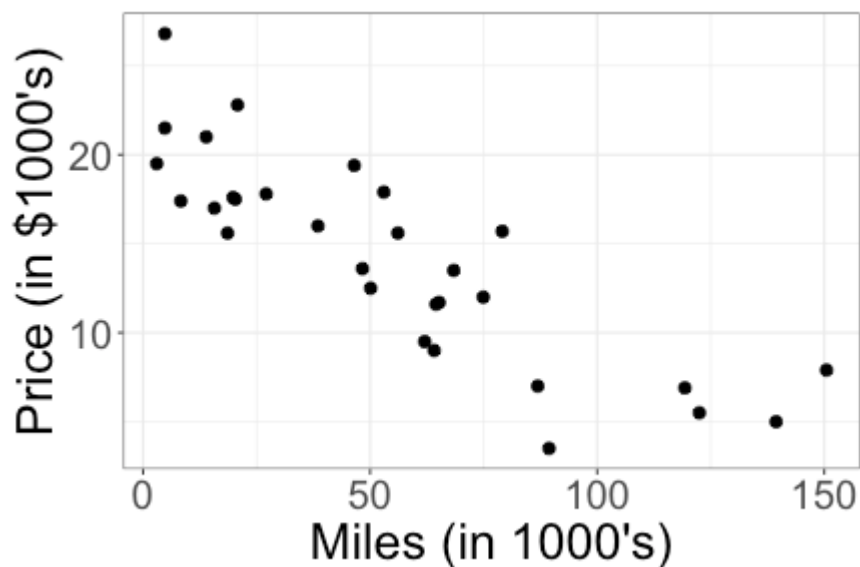


Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

If there is no relationship between mileage and price, what is true about my model?

Translating the question

Is there a relationship between mileage and price for used Honda Accords?

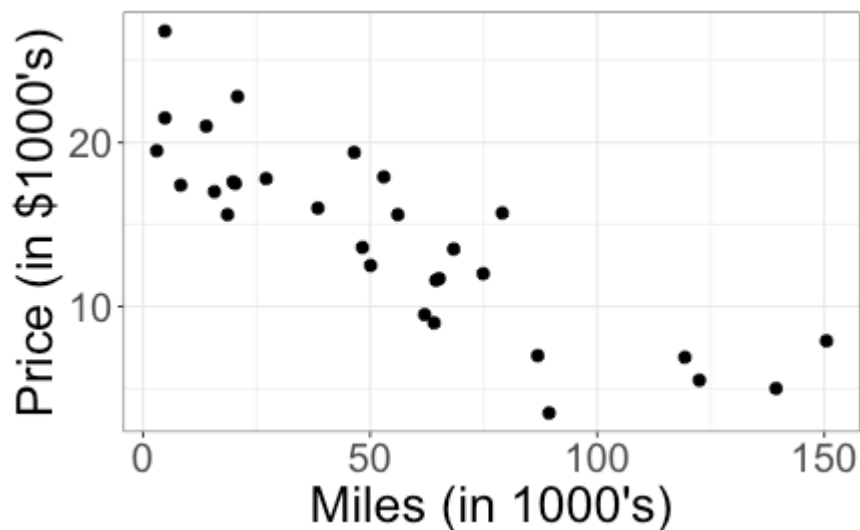


Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Question: Is $\beta_1 = 0$?

Hypothesis testing

$$\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon \quad \text{Is } \beta_1 = 0 ?$$



In statistics, we try to address these kinds of questions with *hypothesis testing*.

Hypothesis testing

What steps does a hypothesis test involve?

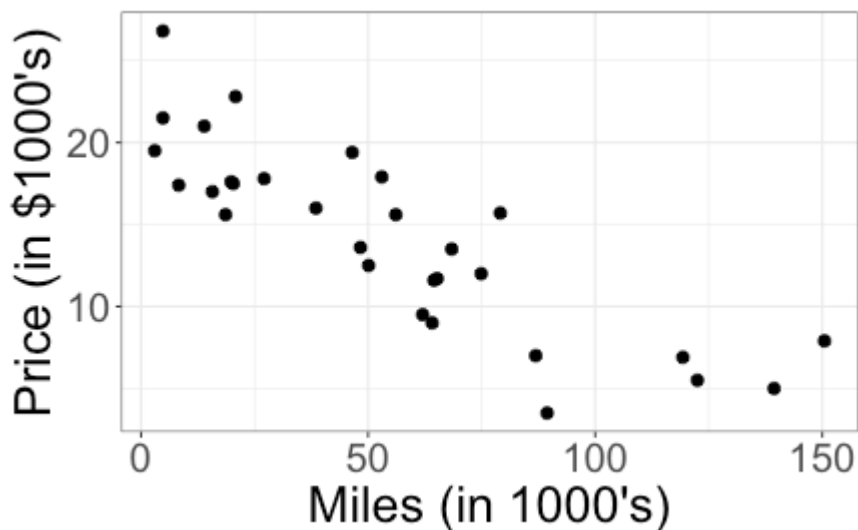
- specify hypotheses: null hypothesis (H_0)
about parameter of interest (e.g. β_1) alternative hypothesis (H_A)
- check conditions (e.g. normality, independence, sample size, etc.)
- choose significance level (α)
- calculate test statistic (e.g. z-stat, t-stat, etc.)
- p-value: probability of "our data or more extreme", under H_0

Hypothesis testing

$$H_0: \beta_1 = 0$$

$$H_A: \beta_1 \neq 0$$

$$\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon \quad \text{Is } \beta_1 = 0 ?$$



In statistics, we try to address these kinds of questions with *hypothesis testing*.

What do you think our null hypothesis (H_0) and alternative hypothesis (H_A) should be?

Specifying hypotheses

Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Question: Is $\beta_1 = 0$?

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

← no $\hat{\beta}$ in any
← of the hypotheses!

- + The *null hypothesis*, H_0 , is that there is no relationship
- + The *alternative hypothesis*, H_A , is that there is a relationship
- + Our hypotheses are stated in terms of one or more model parameters

Specifying hypotheses

Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

What if I want to know whether there is a *positive* relationship between mileage and price?

Specifying hypotheses

Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Hypotheses:

$$H_0 : \beta_1 = 0$$

(two-sided test)

$$H_A : \beta_1 \neq 0$$

What if I want to know whether there is a *positive* relationship between mileage and price?

$$H_0 : \beta_1 = 0$$

(one-sided test)

$$H_A : \beta_1 > 0$$

Estimating the relationship

Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

Estimated line: $\widehat{\text{price}} = 20.8 - 0.12 \text{ miles}$

If $\beta_1 = 0$, how unusual would $\hat{\beta}_1 = -0.12$ be?

Estimating the relationship

Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

Estimated line: $\widehat{\text{price}} = 20.8 - 0.12 \text{ miles}$

If $\beta_1 = 0$, how unusual would $\hat{\beta}_1 = -0.12$ be?

Need to know how $\hat{\beta}_1$ varies from sample to sample. How do we measure the variability of a statistic?

How unusual is the observed slope?

$$\overset{\text{observed slope}}{\downarrow} \quad H_0 : \beta_1 = 0$$

Standardize: $t = \frac{\hat{\beta}_1 - 0}{SE_{\hat{\beta}_1}}$

$\hat{\beta}_1 - 0 \leftarrow \text{value under } H_0$
 $SE_{\hat{\beta}_1} \leftarrow \text{standard error}$

where $SE_{\hat{\beta}_1}$ is the estimated standard error (standard deviation) of $\hat{\beta}_1$

How unusual is the observed slope?

$$H_0 : \beta_1 = 0$$

Standardize: $t = \frac{\hat{\beta}_1 - 0}{SE_{\hat{\beta}_1}}$

Accord data:

response ← explanatory

```
accord_lm <- lm(Price ~ Mileage, data = AccordPrice)
summary(accord_lm)
```

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  20.8096    0.9529   21.84  < 2e-16 ***
## Mileage      -0.1198    0.0141  -8.50 3.06e-09 ***
```

$\hat{\beta}_1$

$SE_{\hat{\beta}_1}$

$$t = \frac{-0.12 - 0}{0.0141} = -8.50$$

How unusual is the observed slope?

$$H_0 : \beta_1 = 0$$

Standardize: $t = \frac{\hat{\beta}_1 - 0}{SE_{\hat{\beta}_1}}$

Accord data:

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  20.8096    0.9529   21.84  < 2e-16 ***
## Mileage      -0.1198    0.0141   -8.50  3.06e-09 ***
```

What does a value of $t = -8.50$ mean?

observed slope (-0.12) is 8.50 SDs below
value under H_0 (0)

How unusual is the observed slope?

Accord data:

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  20.8096      0.9529   21.84  < 2e-16 ***
## Mileage      -0.1198      0.0141   -8.50  3.06e-09 ***
```

The observed value of $\hat{\beta}_1$ (-0.12) is 8.5 standard deviations below the value of β_1 hypothesized in $H_0 : \beta_1 = 0$

Is 8.5 standard deviations a lot?

A lot of SDs! $\Rightarrow \hat{\beta}_1$ is unusual if $\beta_1 = 0$

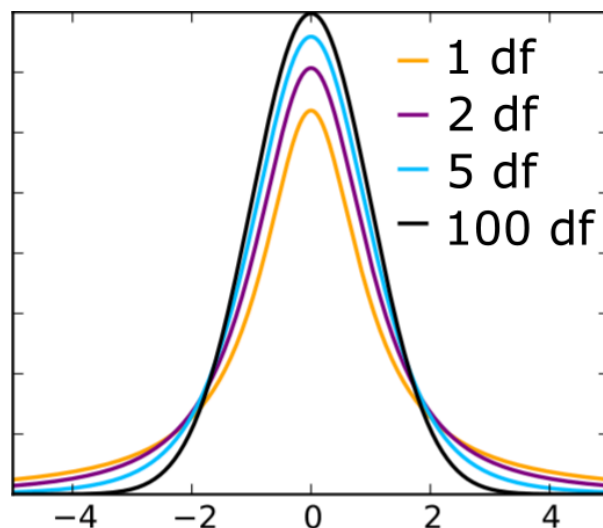
How unusual is the observed slope?

$$t = \frac{\hat{\beta}_1 - 0}{SE_{\hat{\beta}_1}}$$

observations in data
parameters in model (β_0, β_1)

If $\beta_1 = 0$, then t follows a t -distribution with $n - 2$ degrees of freedom (df)

t-distributions



as $df \rightarrow \infty$,
 t distribution
approaches
 $N(0, 1)$

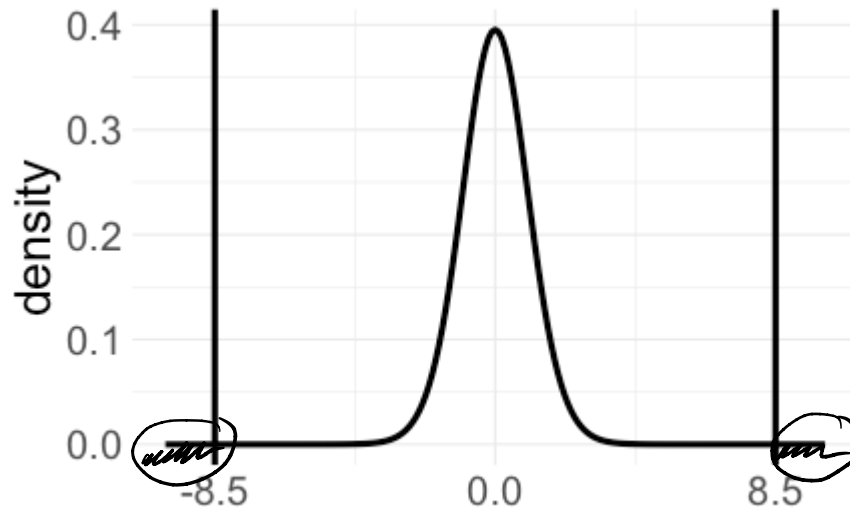
How unusual is the observed slope?

$$H_A: \beta_1 \neq 0$$

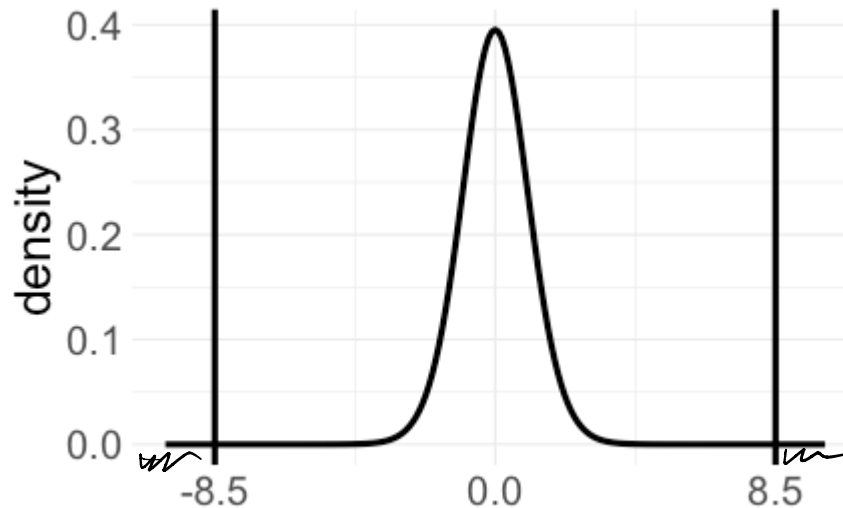
(two-tailed test)

$$t = \frac{-0.12}{0.014} = -8.5, \quad n - 2 = 28$$

t_{28} distribution:



How unusual is the observed slope?

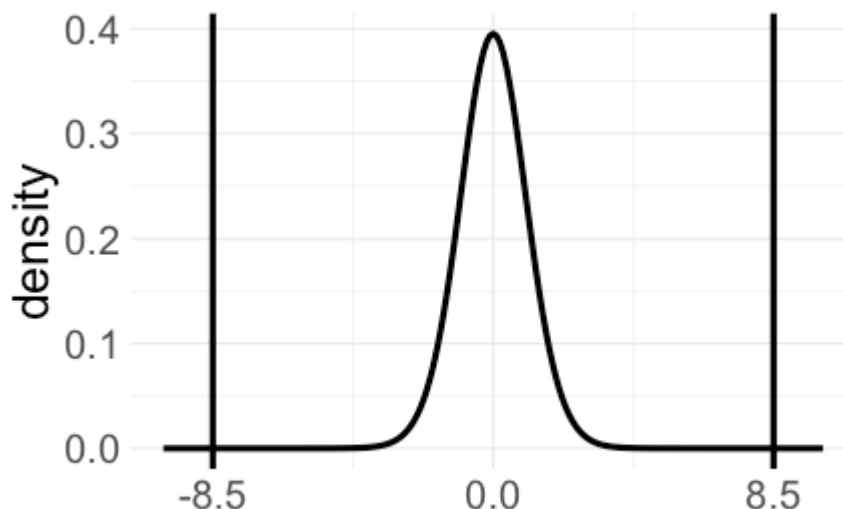


What fraction of the time do we see $t < -8.5$ or $t > 8.5$?

`pt(-8.5, df=28, lower.tail = TRUE) +
pt(8.5, df=28, lower.tail = FALSE)`

look in right tail

How unusual is the observed slope?

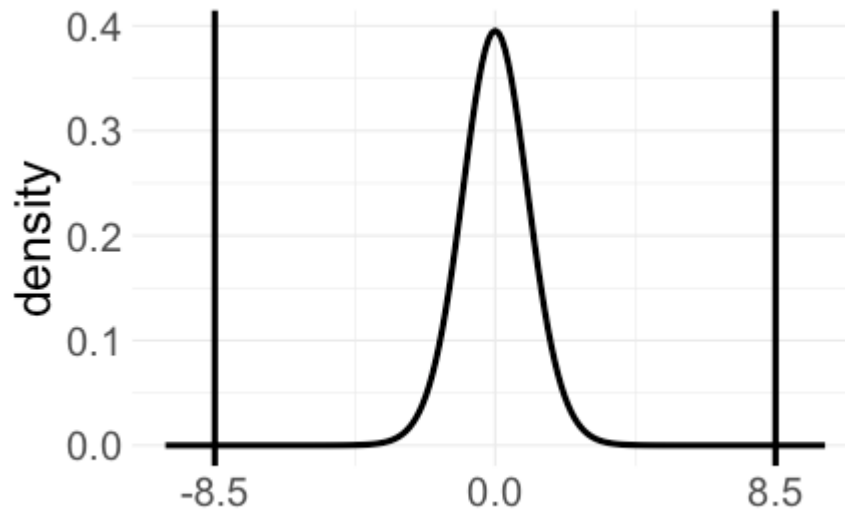


```
pt(-8.5, df=28, lower.tail = TRUE) +  
  pt(8.5, df=28, lower.tail = FALSE)
```

```
## [1] 3.056239e-09
```

If $\beta_1 = 0$, the probability of observing $t < -8.5$ or $t > 8.5$ is approximately 0.00000000003

How unusual is the observed slope?



+ If $\beta_1 = 0$, the probability of observing $t < -8.5$ or $t > 8.5$ is approximately 0.0000000003

+ $p\text{-value} = 0.0000000003$ $\alpha = 0.01$

our $p\text{-value}$ is < 0.01 , so we reject the null hypothesis that $\beta_1 = 0$ (i.e., the hypothesis that there is no relationship between mileage and price)

Putting it together

Is there a relationship between mileage and price for used Honda Accords?

Model: $\text{price} = \beta_0 + \beta_1 \text{mileage} + \varepsilon$

Question: Is $\beta_1 = 0$?

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

Putting it together

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

Estimated line: $\widehat{\text{price}} = 20.8 - 0.12 \text{ mileage}$

Test statistic:
$$t = \frac{\hat{\beta}_1}{SE_{\hat{\beta}_1}} = \frac{-0.12}{0.014} = -8.5$$

p-value: 0.0000000003

Conclusion: The observed slope of -0.12 would be very unusual if there were no relationship between mileage and price for used Honda Accords, so we have strong evidence that there is a relationship between mileage and price.

In R

```
accord_lm <- lm(Price ~ Mileage, data = AccordPrice)
summary(accord_lm)
```

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  20.8096      0.9529   21.84  < 2e-16 ***
## Mileage      -0.1198      0.0141  -8.50 3.06e-09 ***
```

- + R automatically reports p-values for tests of the form

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

↑ p-value for
 $\beta_1 = 0$ vs.
 $\beta_1 \neq 0$

- + *Don't* rely on R output for other tests, such as

- + $H_0 : \beta_1 = 1 \quad H_A : \beta_1 \neq 1$

- + $H_0 : \beta_1 = 0 \quad H_A : \beta_1 > 0$

Class activity

Work on the activity (handout), then we will discuss as a class.

Class activity

$$\text{wing length} = \beta_0 + \beta_1 \text{weight} + \varepsilon$$

Research question: is there a relationship between sparrow weight and wing length?

What are the null and alternative hypotheses?

$H_0: \beta_1 = 0$ (no relationship)

$H_A: \beta_1 \neq 0$ (some relationship)

Class activity

$$\text{wing length} = \beta_0 + \beta_1 \text{weight} + \varepsilon$$

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

Coefficients:

	Estimate	Std. Error
(Intercept)	8.75465	1.39601
Weight	1.31341	0.09756

What is my test statistic for these hypotheses?

$$t = \frac{1.31341 - 0}{0.09756} = 13.46$$

Class activity

$$\text{wing length} = \beta_0 + \beta_1 \text{weight} + \varepsilon$$

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

Coefficients:

	Estimate	Std. Error
(Intercept)	8.75465	1.39601
Weight	1.31341	0.09756

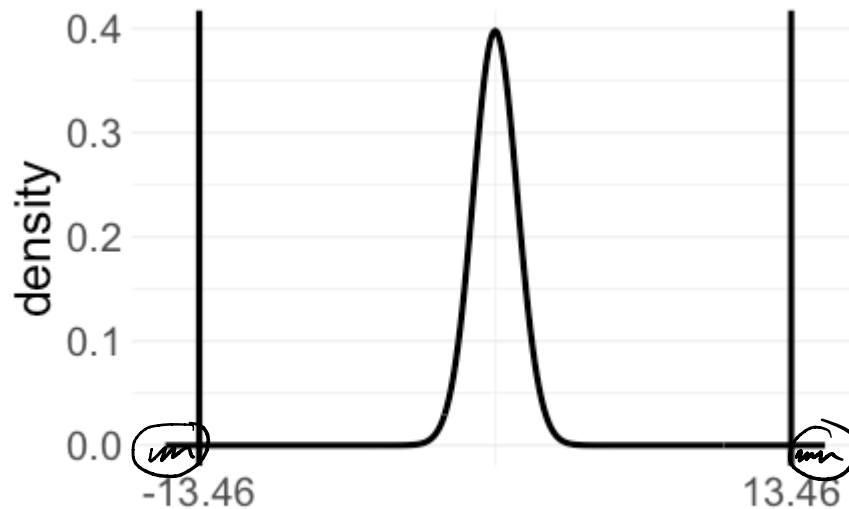
$$t = \frac{\hat{\beta}_1 - 0}{SE_{\hat{\beta}_1}} = 13.46$$

$\hat{\beta}_1$ is 13.46 SDs away
from 0

Is this an "unusual" value of t if H_0 is true? Yes!

Class activity

$$t = 13.46 \quad n - 2 = 114$$



$H_A: \beta_1 \neq 0$

```
pt(-13.46, 114, lower.tail = TRUE) +  
  pt(13.46, 114, lower.tail=FALSE)
```

```
## [1] 2.663963e-25
```

≈ 0